

COURSE MATERIAL

SUBJECT	ENGINEERING PHYSICS (PH23ABS101)
UNIT	I
COURSE	B.TECH
DEPARTMENT	S&H
SEMESTER	II
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1. Course Objectives

The objective of this course are to:

Bridge the gap between the Physics in school at 10+2 level and UG level engineering courses by identifying the importance of the optical phenomenon like interference, diffraction etc, enlightening the periodic arrangement of atoms in crystalline solids and concepts of quantum mechanics, introduce novel concepts of dielectric and magnetic materials, physics of semiconductors.

2. Prerequisites

Students should have knowledge on

1. Wave Optics
2. Crystallography
3. Quantum Mechanics
4. Semiconductors



3. Syllabus

UNIT I

Wave Optics Interference: Introduction - Principle of superposition –Interference of light
Interference in thin films (Reflection Geometry) & applications - Colours in thin films-
Newton's Rings, Determination of wavelength and refractive index.

Diffraction: Introduction - Fresnel and Fraunhofer diffractions - Fraunhofer diffraction due to
single slit, double slit & N-slits (Qualitative) – Diffraction Grating - Dispersive power and
resolving power of Grating (Qualitative).

Polarization: Introduction -Types of polarization - Polarization by reflection, refraction and
Double refraction - Nicol's Prism -Half wave and Quarter wave plates.

4. Courseoutcomes

1. CO 1: Understand the intensity variation of light due to interference, diffraction and polarization.
2. CO 2: Apply the basic concepts of crystal structures and X-ray diffraction to study the behavior of materials for engineering applications.
3. CO 3: Summarize the fundamental properties of dielectric and magnetic materials for engineering applications.
4. CO 4: Analyze the properties of quantum particles to interpret the energy band formation and classification of solids
5. CO 5: Assess the current flow mechanism to understand the transport phenomenon of charge carriers in semiconductors.

5. Co-PO/PSOMapping

APPLIED PHYSICS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3	3												
CO2	3	3												
CO3	3	3												
CO4	3	3												
CO5	3	3												

6. LessonPlan

Lecture No.	Weeks	Topicstobecoved	References
1	1	Introductiontowaveoptics	T1, R1
2		Interference,principleofSuperposition	T1, R1
3		Interferencebydivisionofwavefrontandamplitude	T1, R1
4		Interferenceinthinfilms(reflectedlight)	T1, R1
5		Newton'sRings-theory	T1, R1

6	2	Newton'sRings-DeterminationofWavelength	T1, R1
7		Diffraction, differences between diffraction and interference	T1, R1
8		Fraunhofer Diffraction and Fresnel diffraction	T1, R1
9	3	Single slit	T1, R1
10		Double slit	T1, R1

11		Diffraction grating	T1, R1
12		Gratings spectrum and determination of wavelength	T1, R1
13	4	Polarization-Polarization by double refraction	T1, R1
14		Nicol's Prism	T1, R1
15		Half wave and Quarter wave plate	T1, R1
16		Applications	T1, R1

7. Activity Based Learning

1. Determining thickness of a thin film by two parallel glass plates by using interference method.
2. Determining radius of curvature by using Newton's rings.

8. Lecture Notes

1.1 INTRODUCTION

Have you ever observed that the beautiful colors in soap bubbles, mica and film of oil floating on the surface of water when a sun light fall on them? Did you ever try to find out the reason? This is due to the phenomenon of interference in thin films by reflection of light. This was first explained by Thomas young on basis of Hugen's wave concept and principle of superposition.

1.2 PRINCIPLE OF SUPERPOSITION OF WAVES

When two or more light waves traveling through a medium superimpose one another then the resultant displacement at any point is equal to the algebraic sum of the individual displacements at that point. This is called principle of superposition.

$$\text{i.e.,} \quad \vec{y} = \vec{y}_1 \pm \vec{y}_2 \pm \vec{y}_3 \pm \dots \rightarrow (1.1)$$

where $\vec{y}$ is the resultant displacement and $\vec{y}_1, \vec{y}_2, \vec{y}_3, \dots$ are the displacements of individual waves.

+Vesign is taken when displacements are in same direction.

–Vesign is taken when displacements are in opposite direction.

- (i) If two light waves are in same phase, the resultant displacement is the sum of displacements of two waves. This is called constructive interference.

$$\text{i.e.,} \quad \vec{y} = \vec{y}_1 + \vec{y}_2 \rightarrow (1.2)$$

(ii) If two light waves are out of phase, the resultant displacement is the difference of displacements of two waves. This is called destructive interference

i.e., $\vec{y} = \vec{y}_1 - \vec{y}_2$ →(1.3)

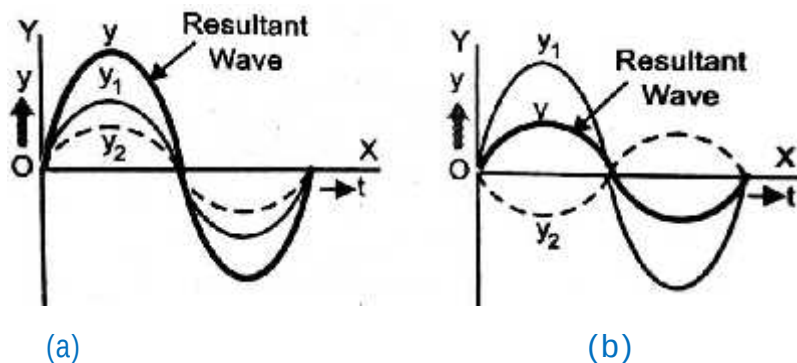


Fig.1.1.1. Superposition of waves

1.3. Interference of light

When two or more light waves traveling through a medium superimpose one another then the resultant intensity at different points of the medium undergoes change from point to point. The change in the intensity of light in a medium from point to point is called interference of light. Thus, the phenomenon of modification or change in uniform distribution of light energy or intensity or amplitude due to the superposition of two or more light waves is called interference.

1.4. Types of Interference

There are two methods used to produce the phenomenon of interference.

1. By Division of Wavefront 2. By Division of Amplitude

1.4.1. Division of Wavefront

In this method, a single (point or narrow slit) source emits wavefronts. A wave front is divided into two parts by using mirrors, biprisms or lenses. These two wave fronts travel different distances and are then superimposed to produce interference.

Examples: Young's double slit Fresnel biprism, Fresnel mirrors, Lloyd's mirror, and lasers, etc.

1.4.2. Division of Amplitude

In this method, the amplitude of the incident beam is divided into two or more parts either by partial reflection or refraction. Thus we have coherent beams produced by division of amplitude. These beams travel different paths and are finally brought together to produce interference.

Examples: Interference in thin films, Newton's rings, Michelson's interferometer, and Fabry-Perot's interferometer, etc.

1.5. Theory of interference

Let us consider a monochromatic source of light S as shown in Fig.1.2. Let S_1 and S_2 are two narrow slits equidistant from S and they act as coherent sources. The waves emitted from S_1 and S_2 superimpose each other at point P on the screen XY .

Let a_1, a_2 be the amplitudes of the waves from S_1 and S_2 .

Let ϕ be the phase difference of the two waves reaching the point P .

Let y_1 and y_2 be the displacements of the two waves, arriving at point P .

$$y_1 = a_1 \sin \omega t \quad \rightarrow (1.4)$$

$$y_2 = a_2 \sin(\omega t + \phi) \quad \rightarrow (1.5)$$

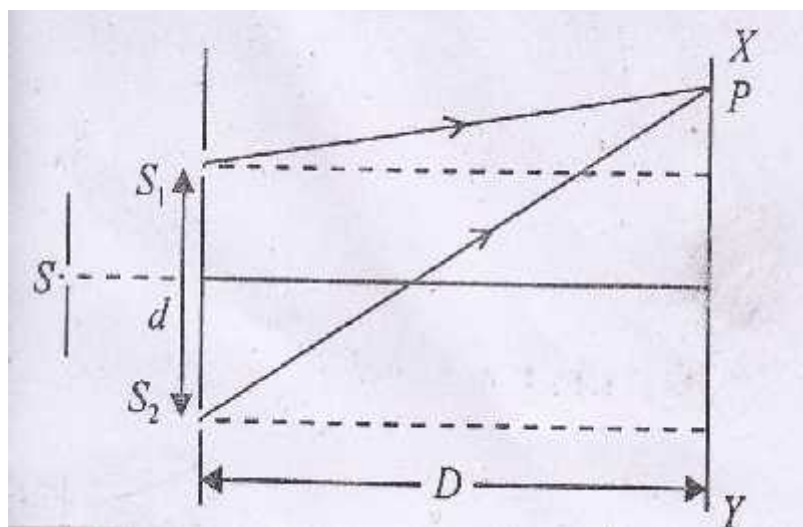


Fig.1.2. Schematic representation of interference due to two slits

According to principle of superposition of waves, the resultant displacement of the waves at P is given by;

$$\begin{aligned} y &= y_1 + y_2 = a_1 \sin \omega t + a_2 \sin(\omega t + \phi) \\ &= a_1 \sin \omega t + a_2 \sin \omega t \cos \phi + a_2 \cos \omega t \sin \phi \\ &= \sin \omega t (a_1 + a_2 \cos \phi) + \cos \omega t (a_2 \sin \phi) \end{aligned} \quad \rightarrow (1.6)$$

$$\text{Let } a_1 + a_2 \cos \phi = A \cos \theta \quad \rightarrow (1.7)$$

$$\text{and } a_2 \sin \phi = A \sin \theta \quad \rightarrow (1.8)$$

Substituting in equation (1.6)

$$\begin{aligned} y &= \sin \omega t A \cos \theta + \cos \omega t A \sin \theta \\ &= A \sin(\omega t + \theta) \end{aligned} \quad \rightarrow (1.9)$$

Squaring equations (1.7) and (1.8), we have $A^2 \cos^2 \theta + A^2 \sin^2 \theta = (a_1 + a_2 \cos \phi)^2 + a_2^2 \sin^2 \phi$

$$\theta + A^2 \sin^2 \theta = (a_1 + a_2 \cos \phi)^2 + a_2^2 \sin^2 \phi$$

$$A^2 (\cos^2 \theta + \sin^2 \theta) = a_1^2 + 2a_1 a_2 \cos \phi + a_2^2 \cos^2 \phi + a_2^2 \sin^2 \phi$$

$$A^2 = a_1^2 + 2a_1 a_2 \cos \phi + a_2^2 (\cos^2 \phi + \sin^2 \phi)$$

$$\phi) A^2 = a_1^2 + 2a_1 a_2 \cos \phi + a_2^2$$

The resultant amplitude of waves is

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi} \rightarrow (1.10)$$

The resultant intensity at point 'P' is given by the square of the amplitude

$$\text{i.e., } I = A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi \rightarrow (1.11)$$

Case 1. Condition for constructive interference

If the two waves are in phase with each other, then they undergo constructive interference producing maximum intensity of light called bright band or fringe.

The resultant intensity will be maximum when $\cos \phi = +1$

$$\text{i.e., } \phi = 0, 2\pi, 4\pi, 6\pi, \dots$$

From equation (1.1), $\phi = n(2\pi)$ where $n = 0, 1, 2, 3, \dots$

$$I_{\max} = a_1^2 + a_2^2 + 2a_1 a_2 \rightarrow (1.12)$$

$$I_{\max} = (a_1 + a_2)^2 \rightarrow (1.13)$$

Let $a_1 = a_2 = a$ (if two light waves are coherent $I_{\max} = I_0$)

$$= (4a)^2 \rightarrow (1.14)$$

Therefore, the intensity is maximum when the phase difference is an integral multiple of 2π or path difference is

an integral multiple of λ because phase difference of 2π corresponds to a path difference of λ .

Path difference Δ

$$= n\lambda \text{ or Phase difference } \phi = 2n\pi \rightarrow (1.15)$$

Case: 2. Condition for destructive interference

If the two waves are out of phase, then they undergo destructive interference producing zero intensity of light called dark band or fringe.

The resultant intensity will be maximum when $\cos \phi = -1$

$$\text{i.e., } \phi = \pi, 3\pi, 5\pi, \dots$$

$$\phi = (2n-1)\pi$$

Where $n=1,2,3,\dots$

From equation (1.1)

$$I_{\min} = a_1^2 + a_2^2 - 2a_1a_2$$

→(1.16)

$$I_{\min} = \frac{(a_1 - a_2)^2}{2}$$

→(1.17)

Let $a_1 = a_2 = a$ (if two light waves are coherent)

$$I_{\min} = 0$$

→(1.18)

Therefore, the intensity is minimum when the phase difference is an odd multiple of π or path difference is an odd multiple of $\lambda/2$ because phase difference of π corresponds to a path difference of $\lambda/2$.

$$\begin{aligned} \text{Path difference } \Delta &= (2n-1)\frac{\lambda}{2} \\ \text{Phase difference } \phi &= (2n-1)\pi \end{aligned}$$

→(1.19)

1.5.1. Intensity distribution graph

The variation of intensity (I) with the phase difference ϕ is as shown in Fig.1.3. The intensity is maximum at bright bands and is minimum at dark bands. From the graph, it is observed that the intensity is uniformly redistributed in the interference pattern.

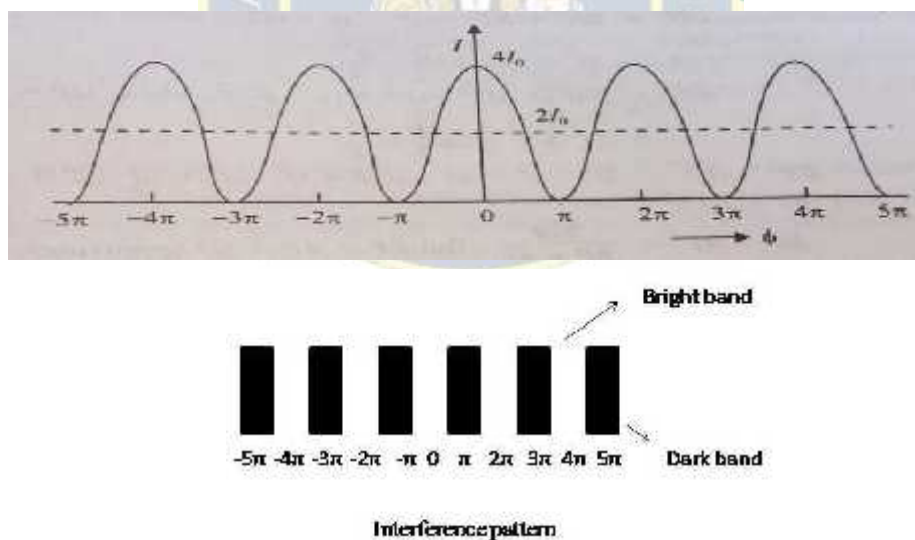


Fig.1.3. Intensity distribution graph

1.6. Conditions for sustained interference

To observe a well-defined interference pattern, the following conditions must be fulfilled.

a) Conditions for Sustained Interference

- To produce interference, we require two coherent light sources. i.e., the two sources must emit light of same frequency or wavelength, same amplitude and in the same phase or with a constant phase.

- ii. The two sources must be perfectly monochromatic, emitting light of a single wavelength or frequency.

b) Conditions for good visibility or observation

- i. The distance between the two sources ($2d$) must be small.
- ii. The distance between the two coherent sources and the screen (D) must be large.

c) Conditions for good contrast

- i. The amplitudes of the two waves must be nearly the same or equal.
- ii. The two sources must be narrow and parallel.
- iii. The two sources must emit light in the same direction.
- iv. To view interference fringes, the background must be dark.

1.7. Interference in thin films

If a light ray is incident on a thin film, a small part of the light is reflected from the top surface and the remaining portion is transmitted into the film. Again, a small part of the transmitted component light is reflected back into the film by the bottom surface and emerges through the top surface. A small portion of the light thus is reflected partially several times in succession within the film. These reflected light waves superimpose with each other, producing interference and forming interference patterns.

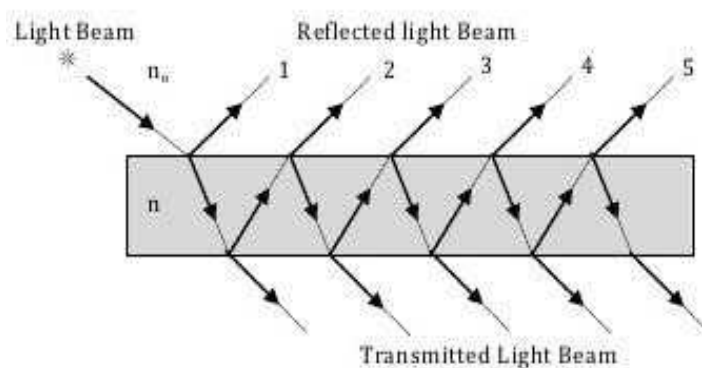


Fig.1.4. Interference in thin films by reflection

1.8. Interference in thin films by reflection of light

When light is incident on a plane parallel thin film, some portion of light gets reflected from the top surface and the remaining portion is transmitted into the film. Again, some portion of the transmitted light is reflected back into the film by the bottom surface and emerges through the top surface. These reflected light waves superimpose with each other, producing interference and forming interference patterns. This is the principle of interference in thin films by reflection of light.

- Let us consider a thin transparent film of uniform thickness t with refractive index μ and is surrounded by air on both sides. Let the refractive index of the air be μ_{air} as shown in Fig 1.5.

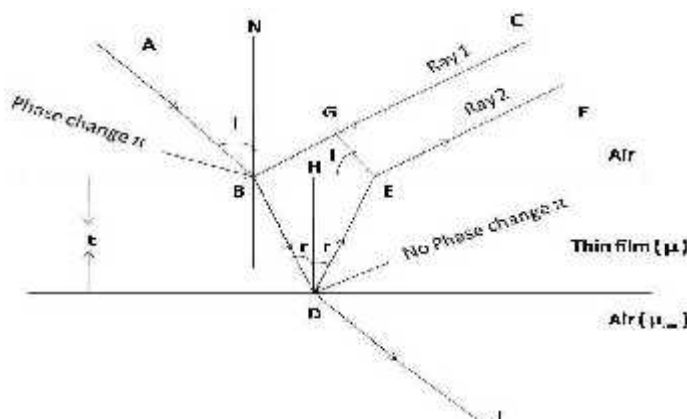


Fig 1.5. Interference in thin films by reflection of light

- When a monochromatic light ray AB is incident on the top surface at an angle i , it is partly reflected along BC and partly refracted into the film along BD making an angle ' r ' with the normal DH and meets the lower surface.
- At D, it is again partly reflected back into the film along DE and emerges along EF which is parallel to the ray BC. The major portion of light refracts into the outer medium along DJ.
- Since the two reflected light rays BC and EF are derived from the incident ray AB, so they act as coherent light rays. These two coherent light rays superimpose and produce interference patterns. Condition of interference depends on the optical path difference between the rays 1 and 2.
- To calculate let us draw normal's from EG to BC and DH to BE and the rays GC and EF equal distances.
- The Optical path difference (OPD) between the rays (1) and (2) is

$$\Delta = \mu \times \text{geometrical path BDE in film} - \mu_{air} \times \text{geometrical path BG in air}$$

$$= \mu(BD + DE) - \mu_{air}(BG) \quad [\text{Since } \Delta = \mu L \text{ and } \mu_{air} = 1]$$

$$= \mu(BD + DE) - BG \quad \rightarrow (1.20)$$

Step 1: Calculation of geometrical path BDE in film
Let us calculate path $BD + DE$ in film,

$$\text{From fig, } \triangle BDH, \cos r = \frac{DH}{BD}$$

$$BD = \frac{t}{\cos r} \quad \rightarrow (1.21)$$

Similarly,

$$\text{From } \triangle EDH, \cos r = \frac{DH}{DE}$$

$$DE = \frac{t}{\cos r} \rightarrow (1.22)$$

$$\therefore BD + DE = \frac{t}{\cos r} + \frac{t}{\cos r} = \frac{2t \cos r}{r} \rightarrow (1.23)$$

Step 2: Calculation of geometrical path BG in film

To calculate BG air, first BE which is equal to (BH + HE) has to be obtained. From $\triangle BDH$

$$\tan r = \frac{BH}{DH} = \frac{BH}{t}$$

$$BH = t \tan r \rightarrow (1.24)$$

Similarly,

$$\text{From } \triangle EDH, \tan r = \frac{HE}{DH} = \frac{HE}{t}$$

$$HE = t \tan r \rightarrow (1.25)$$

$$\therefore BE = BH + HE = t \tan r + t \tan r$$

$$= 2t \tan r \rightarrow (1.26)$$

$$\text{From } \triangle BGE, \sin i = \frac{BG}{BE}$$

$$BG = BE \sin i$$

$$= 2t \tan r \sin i \rightarrow (1.27)$$

$$\text{From Snell's law, } \mu = \frac{\sin i}{\sin r}$$

$$\sin i = \mu \sin r \rightarrow (1.28)$$

From the equations (1.27)

$$\text{and (1.28) } BG = 2t \tan r \mu \sin r$$

$$= 2\mu t \tan r \sin r$$

$$= 2\mu t \frac{\sin r}{\cos r} \sin r$$

$$= 2\mu t \frac{\sin^2 r}{\cos r} \rightarrow (1.29)$$

Substituting the equations (1.23) and (1.29) in equation (1.20); we have

$$\Delta = \frac{2\mu t \cos r}{\cos r} - 2\mu t \frac{\sin^2 r}{\cos r}$$

$$= \frac{2\mu t}{\cos r} (1 - \sin^2 r)$$

$$= \frac{2\mu t \cos^2 r}{\cos r}$$

$$\Delta = 2\mu t \cos r \rightarrow (1.30)$$

This is called Cosine law.

According to Stoke's principle, if a light wave traveling from a rarer medium to a denser medium undergoes a phase change of π or path change $\frac{\lambda}{2}$ (i.e., the wave loses half of wavelength) when it gets reflected at the boundary of a rarer to denser medium.

$$\Delta = 2\mu t \cos r - \frac{\lambda}{2} \quad \rightarrow (1.31)$$

Case 1. Condition for constructive interference or bright band

Bright band occurs when the path difference $\Delta = n\lambda$

$$\rightarrow (1.32)$$

From the above equations (1.31) and (1.32); we have

$$2\mu t \cos r - \frac{\lambda}{2} = n\lambda$$

$$2\mu t \cos r = n\lambda + \frac{\lambda}{2}$$

$$2\mu t \cos r = (2n+1)\lambda \frac{\lambda}{2} \quad \rightarrow (1.33)$$

where $n = 0, 1, 2, 3, \dots$

This is the condition for constructive interference. The film appears bright under this condition.

Case 2. Condition for destructive interference or dark band

$$\text{Dark band occurs when the path difference } \Delta = (2n-1)\lambda \frac{\lambda}{2} \quad \rightarrow (1.34)$$

From the above equations (1.31) and (1.34); we have

$$2\mu t \cos r - \frac{\lambda}{2} = (2n-1)\lambda \frac{\lambda}{2}$$

$$2\mu t \cos r = (2n-1)\lambda + \frac{\lambda}{2}$$

$$2\mu t \cos r = n\lambda \text{ where } n = 0, 1, 2, 3 \quad \rightarrow (1.35)$$

This is the condition for destructive interference. The film appears dark under this condition.

1.9. Interference in Non-uniform thin films - Newton's Rings

Interference in non-uniform thin films by reflection of light was first experimentally observed by Newton.

Definition

When a Plano-convex lens of large radius of curvature is placed convex surface on a plane glass plate, an air film is formed between the lower surface of the Plano-convex lens and the upper surface of the glass plate. The thickness of the air film is zero at point of contact and gradually increases from the point of contact outwards.

If a monochromatic light is allowed to fall normally on this air film, then interference pattern is formed in the form of alternate concentric bright and dark circular rings are formed in the air film due to superposition of reflected light, known as Newton's rings because these rings were discovered by Newton.

Principle

Newton's rings are formed due to interference between the light rays reflected from the top and bottom surfaces of an air film between the glass plate and the Plano-convex lens.

1.9.1 Formation of Newton's rings

- The formation of Newton's rings can be explained with help of Fig 1.6.
- When a light ray (AB) is incident on the system, a part of the light is reflected by the curved surface of the lens (L) and a part is transmitted which is reflected back from the plane surface of the glass plate (P). Ray 1 undergoes no phase change but ray 2 undergoes a phase of π or path change $\frac{\lambda}{2}$ (i.e., the wave loses half of wavelength) when it gets reflected at the boundary of a rarer to denser medium.
- These reflected light rays superimpose with each other producing interference and forming interference patterns in the form of bright and dark circular rings.

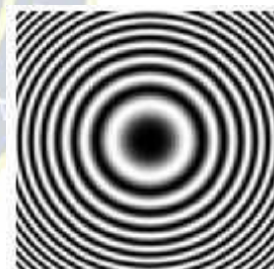
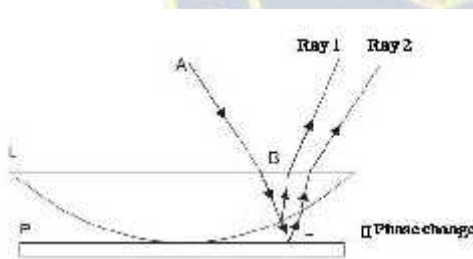


Fig.1.6(a) Formation of Newton's rings

(b) Newton's rings

- We know that, the path difference between two reflected rays in uniform thin film (plane parallel thin film) is

$$\Delta = 2\mu t \cos r - \frac{\lambda}{2} \quad \rightarrow (1.36)$$

- Similarly for non-uniform thin film,

Let refractive index of air film $\mu = 1$ and $r = 0$ for normal incidence.

Thus the path difference

$$\Delta = 2t - \frac{\lambda}{2}$$

$\rightarrow (1.37)$

This is the path difference between two reflected rays in non-uniform thin film.

Case1.Darkcentralspot

At the point of contact 'O' of the lens and glass plate, the thickness of air film is approximately zero.

$$\therefore \text{The path difference } \Delta = 2t - \frac{\lambda}{2} = -\frac{\lambda}{2} \quad \text{i.e., } t \approx 0 \quad \rightarrow (1.38)$$

Thus, at point of contact two waves are out of phase and interference is destructive. Hence a dark spot is formed at the centre.

Case2.Condition for bright band

Bright band occurs when the path difference $\Delta = n\lambda$

$$\rightarrow (1.39)$$

From equations (1.37) and (1.39), we have

$$\begin{aligned} 2t - \frac{\lambda}{2} &= n\lambda \\ 2t &= n\lambda + \frac{\lambda}{2} \\ 2t &= (2n+1)\frac{\lambda}{2} \end{aligned} \quad \rightarrow (1.40)$$

where $n = 0, 1, 2, 3, \dots$

This is the condition for maxima. The film appears bright under this condition.

Case3.Condition for dark band

$$\text{Dark band occurs when the path difference } \Delta = (2n-1)\frac{\lambda}{2} \quad \rightarrow (1.41)$$

From the above equations (1.37) and (1.41), we have

$$\begin{aligned} 2t - \frac{\lambda}{2} &= (2n-1)\frac{\lambda}{2} \\ 2t &= (2n-1)\frac{\lambda}{2} + \frac{\lambda}{2} \\ 2t &= n\lambda \end{aligned} \quad \rightarrow (1.42)$$

where $n = 0, 1, 2, 3, \dots$

This is the condition for minima. The film appears dark under this condition.

1.9.2. Experimental Arrangement

- The experimental arrangement for producing Newton's rings is as shown in Fig 1.7.
- Keep the convex surface of the Plano-convex lens over the plane glass plate and arrange glass plate at an angle of 45° over the base set.
- Switch on the monochromatic light source 'S' (Sodium vapor lamp) and its focus is on the Double convex lens (L). This sends a parallel beam of light. This beam of light falls on the glass plate B at 45° .
- The glass plate 'G' reflects a part of light towards the air film enclosed by the Plano-convex lens and the plane glass plate.

- A part of the light is reflected by the curved surface of the Plano-convex lens (P) and a part is transmitted which is reflected back from the plane surface of the plane glass plate (E).
- These reflected light rays superimpose with each other producing interference and forming interference patterns in the form of bright dark circular rings. These rings are seen with a microscope (M) focused on the air film.

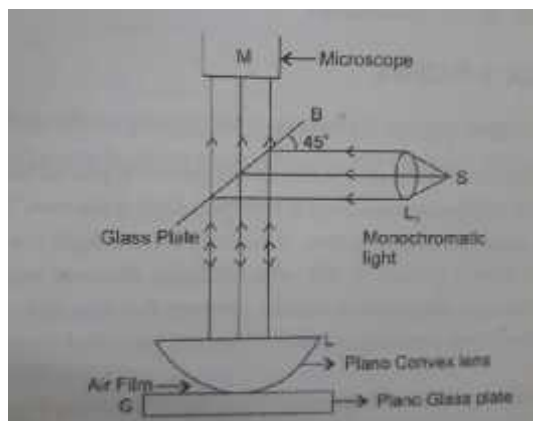


Fig.1.7. The experimental arrangement of Newton's rings

1.9.3 Theory of Newton's rings

To find the diameters (radii) of dark and bright rings, let 'L' be a Plano convex lens placed on a glass plate P. The convex surface of the lens is the part of a spherical surface with center at C as shown in Fig.1.8.

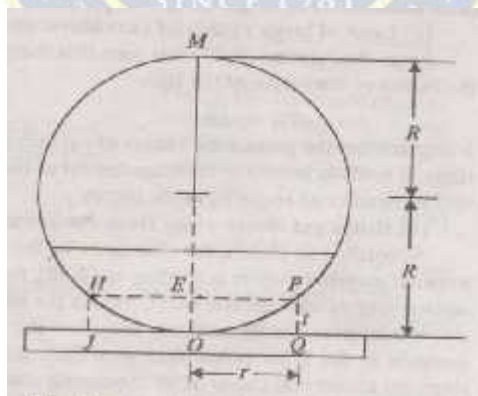


Fig.1.8. Theory of Newton's rings

Case 1. Diameters (Radii) of the dark rings

Let 'R' be the radius of curvature of the lens. Let a

dark ring be located at the point Q such that $PQ = t$. The radius of the ring at Q is $OQ = r$.

By the theorem of intersecting chords,

$$EP \times HE = OE \times EM$$

$$\rightarrow (1.43)$$

But $EP = OQ = HE = r$; $OE = PQ = t$ and EM

$$= OM - OE = 2R - t$$

From equation (1.43) $r^2 = (2R - t)^2 - t^2$

$$r^2 = 4Rt - t^2$$

$$(or) \quad r^2 = 2Rt - t^2$$

As $2Rt \gg t^2$; t^2

can be neglected. $r^2 = 2Rt$

Thus, the radius of the n th dark ring will be given by $r_n^2 = 2Rt$

→

(1.44) We know that, the condition of the dark ring is $2t = n\lambda$

→ (1.45)

From the above equations (1.44) and (1.45)

$$r_n^2 = n\lambda R$$

$$r_n = \sqrt{n\lambda R}$$

→ (1.46)

This is the condition for radii of the dark rings.

The diameter of the dark ring is therefore given by $D_n = 2r_n$

$$D_n = 2\sqrt{n\lambda R}$$

→ (1.47)

∴ The radii (diameters) of the dark rings is directly proportional to

(i) $\sqrt{n}$ (natural number)

(ii) $\sqrt{\lambda}$

(iii) $\sqrt{R}$

Case 2. Diameters (Radii) of the bright rings

Let us now suppose that a bright ring be located at the point Q.

Therefore, the radius of the n th bright ring will be given by $r_n^2 = 2Rt$

→ (1.48)

We know that, the condition of the bright band is $2t = (2n + 1)\frac{\lambda}{2}$

→ (1.49)

From the above equations (1.48) and (1.49)

$$r_n^2 = (2n + 1)\frac{\lambda}{2} R = (n + \frac{1}{2})\lambda R$$

$$r_n = \sqrt{(n + \frac{1}{2})\lambda R}$$

→ (1.50)

This is the condition for radii of the bright rings.

The diameter of the bright ring is therefore given by $D_n = 2r_n$

$$D_n = 2\sqrt{(n + \frac{1}{2})\lambda R}$$

→ (1.51)

- ∴ The radii (diameters) of the bright rings is directly proportional to (i) $\sqrt{n+1}$
 (ii) $\sqrt{\lambda}$
 (iii) $\sqrt{R}$

1.10. Applications

The theory of Newton's rings can be used

- To determine the wavelength of monochromatic light
- To determine the radius of curvature of Plano-convex lens and
- To determine the refractive index of a given liquid.

1.10.1. Determination of wavelength of monochromatic light source.

Let 'R' be the radius of curvature of a Plano-

convex lens, λ be the wavelength of monochromatic light source.

Let D_m and D_n are the diameters of m^{th} and n^{th} dark rings respectively.

Then,

$$\begin{aligned} D_m^2 &= 4m\lambda R, \\ D_n^2 &= 4n\lambda R, \\ \text{and } D_n^2 - D_m^2 &= 4n\lambda R - 4m\lambda R \\ &= 4(n-m)\lambda R \end{aligned} \rightarrow (1.52)$$

$$\lambda = \frac{D_n^2 - D_m^2}{4(n-m)R}$$

A graph is drawn with the number of fringes on the x-axis and the square of the diameter of the ring (D^2) on the y-axis. The graph is a straight line passing through the origin as shown in Fig. 1.9. From the graph the values of D_m^2 and D_n^2 corresponding to n^{th} and m^{th} rings are found. From the graph, the slope is calculated.

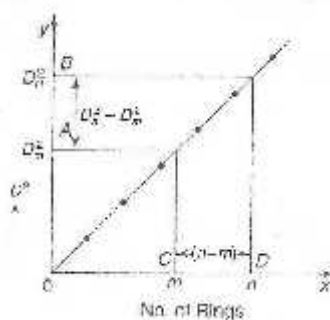


Fig. 1.9. A graph between (D^2) and number of fringes

h, the slope

$$\frac{D_n^2 - D_m^2}{(n-m)} = \frac{AB}{CD} \rightarrow (1.53)$$

The radius of curvature 'R' of the Plano-convex lens is found by Boy's method or spherometer. Substituting the values of R and $D_n^2 - D_m^2$ in the above formula (1.53) μ can be calculated.

1.10.2. Determination of refractive index of a given liquid

To find the refractive index of a given liquid, the Plano convex lens and glass plate set up is placed in a small container 'C'. The transparent liquid of refractive index ' μ ' is introduced between the lens L and the glass plate 'G' as shown in Fig 1.10. Then a film of liquid is formed between the lens and glass plate. The diameters of mth and nth dark rings are determined with the help of travelling microscope.

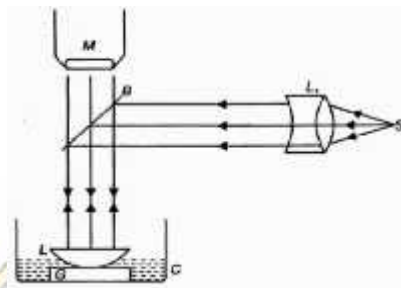


Fig.1.10. Determination of refractive index of a given liquid

We know that,

The diameters of mth and nth dark rings in the air film are given by

$$\begin{aligned} D_m^2 &= 4m\lambda R, D_n^2 = 4n\lambda R, \\ \text{and } D_n^2 - D_m^2 &= 4n\lambda R - 4m\lambda R \\ &= 4(n-m)\lambda R \end{aligned}$$

→ (1.54)

With liquid film, the diameters of mth and nth dark rings are determined;

$$\begin{aligned} D_m^2 &= \frac{4m\lambda R}{\mu} \\ D_n^2 &= \frac{4n\lambda R}{\mu} \\ \text{and } D_n^2 - D_m^2 &= \frac{4n\lambda R}{\mu} - \frac{4m\lambda R}{\mu} \\ &= \frac{4\lambda R(n-m)}{\mu} \end{aligned} \quad \rightarrow (1.55)$$

From the above equations (1.54) and (1.55), $D_n^2 - D_m^2 = \frac{4\lambda R(n-m)}{\mu}$

$\mu = \frac{D_n^2 - D_m^2}{D_n^2 - D_m^2} \cdot \frac{4\lambda R(n-m)}{4\lambda R(n-m)}$. Using the above formula μ can be calculated.

1.11. Applications of Interference

The phenomenon of interference can be used in various applications; some of the applications are given below.

1. It is used in anti-reflecting coatings.
2. It is used to determine the thickness of a thin film coating.
3. It is used in testing of flatness of surfaces.
4. It is used in interference filters to transmit a very narrow range of wavelengths.

2.0. DIFFRACTION

2.1. Introduction

Diffraction is one of the natural phenomena. The effect of diffraction is usually seen in everyday life. For example, the iridescent colors of peacock feathers, butterfly wings and some other insects, rainbow-like diffraction are produced by CD or DVD.

The English word diffraction was coined by the 'diffractus' which means to spread out or to break out. This phenomenon was first discovered by Francesco Grimaldi.

When light waves encounter an obstacle, they bend at the corners or edges of the obstacle and spread into the geometrical shadow of an obstacle. This phenomenon is known as diffraction.

(or)

The phenomenon of redistribution of light intensity due to the superposition of secondary wavelets from the same wave front is called diffraction.

The effect of diffraction depends on the size of the obstacle (d) and the wavelength (λ) of light waves as illustrated in Figure 1.10.

Light waves are very small in wavelength, i.e., from 4×10^{-7} m to 7×10^{-7} m. If the size of opening or obstacle is near to this limit, only then we can observe the phenomenon of diffraction.

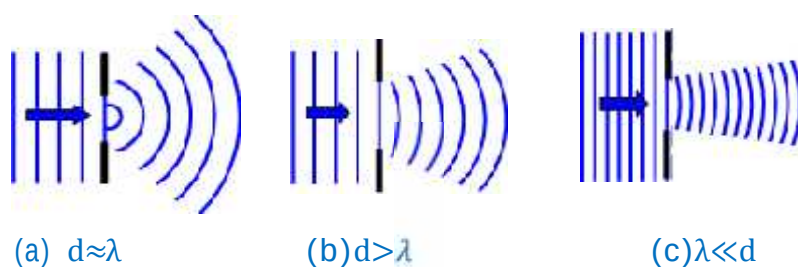


Figure 1.11. Illustration of diffraction phenomenon

From Fig.1.11, it is observed that

- The effect of diffraction is more noticeable if the size of the obstacle is equal to the wavelength of light (i.e. $d \approx \lambda$).
- The effect of diffraction is not noticeable if the size of the obstacle is greater than the wavelength of light (i.e. $d > \lambda$).
- The effect of diffraction is less noticeable if the size of the obstacle is smaller than the wavelength of light (i.e. $\lambda \ll d$).

2.2. Conditions for diffraction

To observe diffraction

- Light must be monochromatic and
- The wavelength of light must be comparable to the obstacle. ($\lambda \approx d$)

2.3. Differences between interference and diffraction

S.No	Interference	Diffraction
1	It arises due to the superposition of waves from the two coherent sources.	It arises due to the superposition of secondary wavelets from the different parts of the same wavefront.
2	It is formed due to the uniform distribution of light intensity.	It is formed due to the non-uniform distribution of light intensity.
3	In interference pattern, all bright fringes are of the same intensity.	In diffraction pattern, all bright fringes are not of the same intensity.
4	The intensity of all dark fringes is zero.	The intensity of all dark fringes is not zero.
5	In interference pattern, the width of fringes is equal.	In diffraction pattern, the width of fringes is not equal.

2.4. Types of Diffraction

There are two types of diffraction.

- Fresnel diffraction and
- Fraunhofer diffraction.

(a) Fresnel diffraction

- In this class of diffraction, the source of light and the screen are finite from the diffraction slit (or) aperture.
- No double convex lenses (or) mirrors are used to observe diffraction effect.
- The diffraction pattern is the image of the slit.
- The centre of diffraction pattern is either bright (or) dark.
- The incident wave front is spherical (or) cylindrical.
- It is very easy to observe experimentally and complex mathematically.

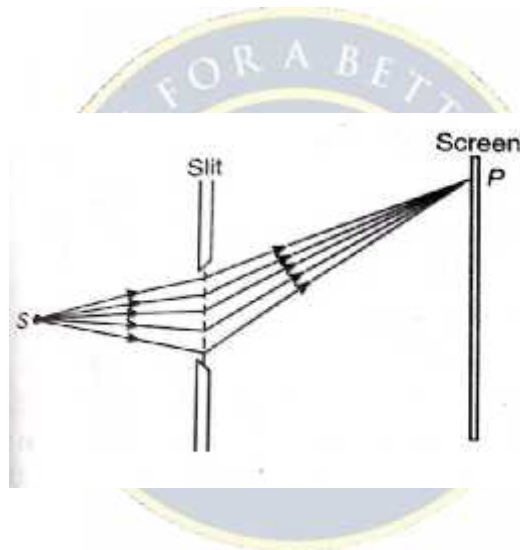


Fig.1.12. Fresnel diffraction

(b) Fraunhofer diffraction

- In this class of diffraction, the source of light and the screen are at infinite distance from the diffraction slit (or) aperture.
- Two double convex lenses (or) mirrors are used to observe diffraction effect.
- The incident wave front is plane.
- The diffraction pattern is the image of the source of itself.
- The centre of the diffraction pattern is bright.
- It is very simple and easy to observe both experimentally and mathematically.

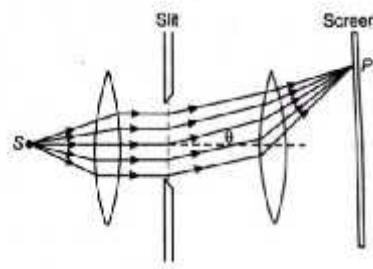


Fig.1.13.Fraunhoferdiffraction

2.5 .Fraunhoferdiffractionduetosingleslit

ThearrangementoffraunhoferdiffractionduetosingleslitasshowninFig1.14.

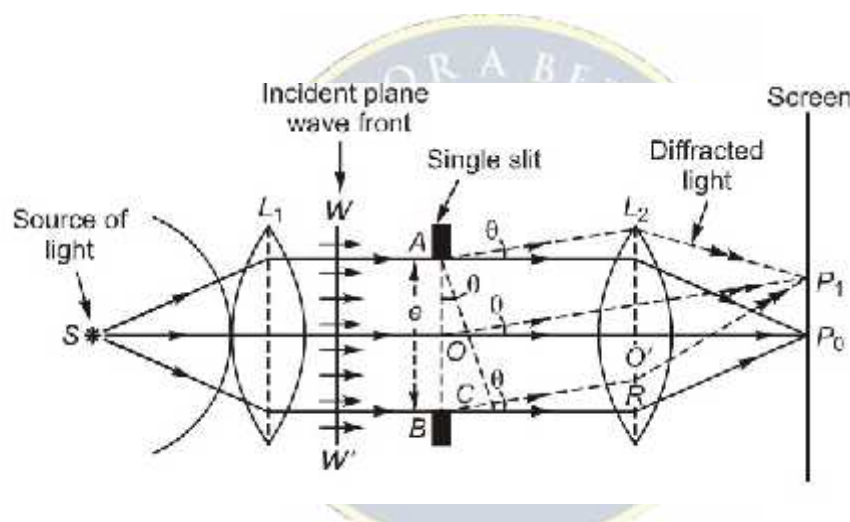


Fig1.14.Fraunhoferdiffractionduetosingleslit

- In Fig S is a point source of monochromatic light with wave length λ and AB is thesingleslitofwidth“e”
- LetWW'betheplanwavefrontandisincidentonthesingleslitAB.
- According to Hygen's principle, each point on wave front is a source of freshdisturbance called secondary wavelet that spread out to right angles in alldirections.
Withoutdiffraction($\theta=0$)
- Let us consider the wave front WW'travelling along OP_0 through the slit AB withoutdiffraction and focus at P_0 by using lens L_2 . Since all these wave fronts have samephaseandnopathdifferencebetweenthem.Theyundergoconstructiveinterferenceandproducingmaximumintensitycalledcentralmaximumorprincipal maximum.

With diffraction ($\theta \neq 0$)

- Let us consider the plane wave front WW' (secondary wavelets) travelling along OP_1 and diffract at an angle θ and focused at P_1 . The intensity at P_1 is either maximum or minimum depending up on the path difference between the two wavelets (A and B). To calculate path difference let us draw a normal AC to BR as shown in fig.

From fig, ΔABC , $\sin \theta = \frac{BC}{AB}$

$$BC = AB \sin \theta$$

$\therefore$ The Path difference $BC = e \sin \theta \rightarrow (1.56)$

And corresponding phase difference $= \frac{2\pi}{\lambda} \times \text{path difference}$

$$= \frac{2\pi}{\lambda} \times e \sin \theta \rightarrow (1.57)$$

Let us consider the width of slit AB is divided into n parts and amplitude of each part is " a ." Therefore, the phase difference between any two successive parts is

$$\frac{1}{n} \times \text{path difference} = \frac{1}{n} \times \text{path difference}$$

$$\frac{1}{n} \times \text{phase difference} = \frac{1}{n} \left[\frac{2\pi}{\lambda} \times e \sin \theta \right] = d \rightarrow (1.58)$$

By using the method of vector addition of amplitudes,

The resultant amplitude is given by $R = \frac{a \sin \frac{n\alpha}{2}}{\sin \frac{\alpha}{2}} \rightarrow (1.59)$

Substituting equation (1.58) in equation (1.59)

$$R = \frac{a \sin n \times \left(\frac{\lambda}{2} \times \frac{2\pi \times e \sin \theta}{\lambda} \right)}{\sin \times \left(\frac{\lambda}{2} \right)} = \frac{a \sin \left(\frac{\pi e \sin \theta}{\lambda} \right)}{\sin \left(\frac{\pi e \sin \theta}{\lambda} \right)}$$

$$R = \frac{a \sin \alpha}{\sin \frac{\alpha}{n}}$$

where $\alpha = \frac{\pi e \sin \theta}{\lambda} \rightarrow (1.60)$

Since $\sin \frac{\alpha}{n}$ is very small, $\sin \frac{\alpha}{n} \approx \frac{\alpha}{n} = \frac{a \sin \alpha}{n} = \frac{na \sin \alpha}{n} = \frac{\alpha}{n} \quad [\because na = A]$

$$\boxed{R = \frac{a \sin \alpha}{\frac{\alpha}{n}}} \rightarrow (1.61)$$

This equation represents the resultant amplitude of the wave fronts at P_1 .

And intensity $I = R^2$

$$I = A^2 \left(\frac{\sin \alpha^2}{\alpha} \right)$$

→(1.62)

$$I = A^2 \left(\frac{\sin \alpha^2}{\alpha} \right)$$

Case 1. Principle maximum or Central maximum at P_0

The resultant amplitude (R) can be expressed in ascending power of α ,

$$R = \frac{A}{\alpha} \left[\alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \frac{\alpha^7}{7!} + \dots \right]$$

$$= \frac{A}{\alpha} \left[1 - \frac{\alpha^2}{3!} + \frac{\alpha^4}{5!} - \frac{\alpha^6}{7!} + \dots \right]$$

→(1.63)

Intensity at P_0 will be maximum when the values of R is maximum.

For maximum value of R the negative terms must be vanished in equation (1.63)

i.e., $\alpha = 0$ [From the equation (1.60)]

$$\frac{\pi e \sin \theta}{\lambda} = 0$$

$$\pi e \sin \theta = 0 \text{ Si}$$

$$n\theta = 0$$

$$\theta = 0$$

Thus we can say that, the condition $\theta = 0$ means that the wave fronts passing through the slit AB without diffraction and focus at P_0 . Since all these wave fronts have same phase and no path difference between them. Therefore, they undergo constructive interference and producing maximum intensity called central maximum or principal maximum (i.e., $I_{\max} = R^2 = A^2 = I_0$).

Case 2. Minimum intensity

positions Intensity will be minimum when $\sin$

$$\alpha = 0. \text{ i.e., } \alpha = \pm m\pi$$

Where $m = 1, 2, 3, \dots$

$$\frac{\pi e \sin \theta}{\lambda} = \pm m\pi$$

$$e \sin \theta = \pm m\lambda$$

→(1.64)

This is the condition for minimum intensity. Thus, we obtain minimum intensity positions either side of the principle maximum.

Case.3.Maximumintensitypositions(or)Secondarymaximum

In addition to principle maximum, there exist weak secondary maximum between equally spaced minima. These secondary maximum intensity positions can be obtained by differentiating the expression of intensity with respect to " α " and equating to zero.

$$I = A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

$$\frac{dI}{d\alpha} = \frac{d}{d\alpha} \left[A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \right] = 0$$

$$= A^2 \frac{d}{d\alpha} \left(\frac{\sin \alpha}{\alpha} \right)^2 = 0$$

$$\left[\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \right]$$

$$= A^2 \cdot \frac{2 \sin \alpha}{\alpha} \cdot \frac{d}{d\alpha} \left(\frac{\sin \alpha}{\alpha} \right) = 0$$

$$= A^2 \cdot \frac{2 \sin \alpha}{\alpha} \times \frac{\alpha \cos \alpha - \sin \alpha}{\alpha^2} = 0 \quad \rightarrow (1.65)$$

In the above equation (1.65) either $\sin \alpha = 0$ (or) $\alpha \cos \alpha - \sin \alpha = 0$

We know that, $\sin \alpha = 0$ gives the minimum intensity positions.

Thus $\alpha \cos \alpha - \sin \alpha = 0$ gives the maximum intensity positions.

i. e., $\alpha \cos \alpha - \sin \alpha = 0$

$\cos \alpha = \sin \alpha$

$$\alpha = \tan \alpha$$

$\rightarrow (1.66)$

The value of α satisfying the above equations are obtained graphically by plotting the curves $y = \alpha$ and $y = \tan \alpha$ on the same graph as shown in Fig 1.15.

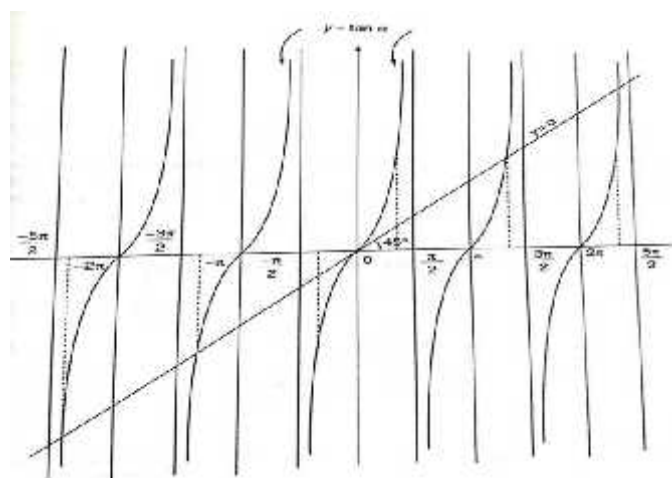


Fig1.15.The plotsof $y = \alpha$ and $y = \tan \alpha$

The points of intersection of the two curves give the value of α which satisfying the above equation (1.66). The points of intersection are $\alpha = 0, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots$

The first value of α gives the principle maximum and remaining values of α give the secondary maxima.

The values are $\alpha = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots$

$$\alpha = \pm (2m+1) \frac{\pi}{2} \rightarrow (1.67)$$

where $m = 1, 2, 3, 4, \dots$

$\therefore$ From the equations (1.60) & (1.67)

$$\frac{\pi \sin \theta}{\lambda} = \pm (2m+1) \frac{\pi}{2}$$

$$\sin \theta = \pm (2m+1) \frac{\lambda}{2a}$$

$\rightarrow (1.68)$

This is the condition for maxima.

(i) For the principle maximum $\alpha = 0$

$\therefore$ The

$$\text{intensity } I = R^2 = A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

$$R^2 = A^2 = I_0$$

(ii) For the secondary maxima $\alpha = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots$

The intensity of first secondary maxima, $\alpha = \pm \frac{3\pi}{2}$

$$\therefore \text{Intensity } (I_1) = A^2 \left(\frac{\sin \left(\frac{3\pi}{2} \right)}{\frac{3\pi}{2}} \right)^2$$

$$= \frac{4}{9} A^2$$

$$= \frac{A^2}{2.25}$$

$$I_1 = \frac{I_0}{2.25}$$

$$\frac{I_1}{I_0} = \frac{1}{2.25} = 4.4\%$$

$\therefore$ The intensity of first secondary maxima is about $\frac{1}{2.25}$ of intensity of the principle maximum.

$$\text{Intensity of first secondary maxima is } \frac{1}{2.25} \text{ of } I_0$$

The intensity of second secondary maxima $\alpha = \pm \frac{5\pi}{2}$

$$I_2 = A^2 \left(\frac{\sin \left(\frac{5\pi}{2} \right)}{\frac{5\pi}{2}} \right)^2$$

$$= A^2 \left(\frac{4}{25\pi^2} \right)$$

$$\frac{I}{I_0} = \frac{A^2}{62} = \frac{I_0}{62}$$

$$\frac{I_2}{I_0} = \frac{1}{62} = 1.61\%$$

∴ The intensity of second secondary maxima is about $\frac{1}{62}$ of intensity of principle maxima.

Secondary max	% of I_0
i.e. I_2	1.61%

∴ In the same way the intensity of secondary maxima goes on decreasing very rapidly.

Intensity distribution graph

A graph is plotted between intensity of light (I) and α is shown in Fig 1.16.

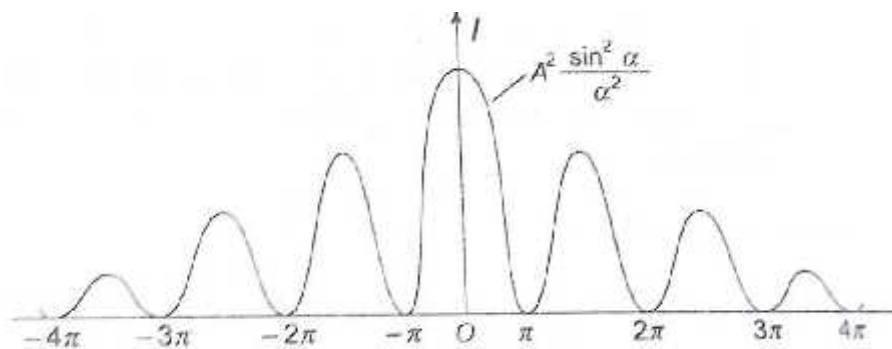


Fig 1.16. The intensity of diffraction pattern due to single slit.

From the graph the following conclusions can be observed,

- At $\alpha = 0$, the intensity is very high called principle maximum.
- At $\alpha = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots$, the intensity is maximum called secondary maxima.
- At $\alpha = \pm \pi, \pm 2\pi, \pm 3\pi, \dots$, the intensity is minimum.
- The most of the light is concentrated in the principle maximum & the intensity of secondary maximum decreasing on either sides of principle maximum as shown in Fig 1.16.

2.6. Fraunhofer diffraction due to double slit

The arrangement of Fraunhofer diffraction due to double slit as shown in Fig 1.17.

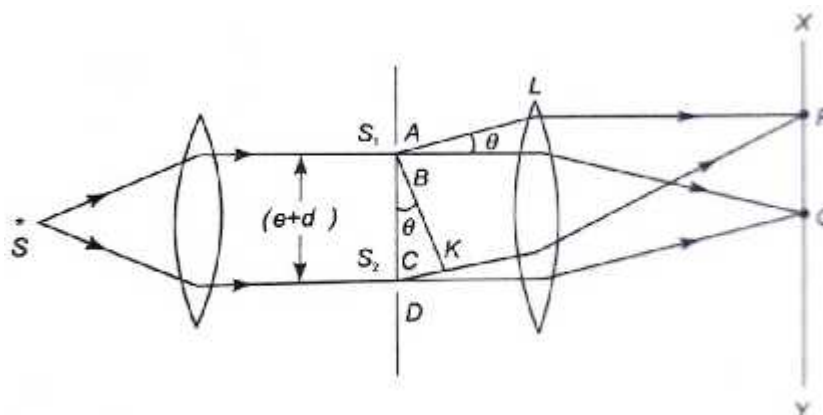


Fig 1.17. Fraunhofer diffraction due to double slit

- In Figure 1.16, S is a point source of monochromatic light with wavelength λ and S_1 and S_2 are the two slits of same width 'e' and separated by a distance 'd'.
- Let WW' be the plane wave front and is incident on the two slits S_1 and S_2 .
- According to Huygen's principle, each point on wave front is a source of fresh disturbance called secondary wavelet that spread out to right angles in all directions.

Without diffraction ($\theta = 0$)

- Let us consider the plane wave front ww' travelling along OP_0 , through the slits S_1 and S_2 without diffraction ($\theta = 0^\circ$) and focused at P_0 by using lens L_2 . Since all these wave fronts have same phase and no path difference between them. So they undergo constructive interference and producing maximum intensity at P_0 called central maximum or principle maximum.

With diffraction ($\theta \neq 0$)

- Let us consider the plane wave front ww' (secondary wavelets) travelling along OP_1 through the slits S_1 and S_2 and diffract at an angle θ with respect to the normal and focused at P_1 . The intensity at P_1 is the combination of both interference and diffraction.
- Interference: It is due to the superposition of secondary diffracted wavelets from the two slits S_1 and S_2 .

- Diffraction: It is due to the single slit, the intensity at P_1 either maximum or minimum depending up on the path difference between secondary wavelets s_1 and s_2 . From $\Delta s_1 s_2 k$,

$$\sin \theta = \frac{s_2 k}{s_1 s_2}$$

$$s_1 s_2 \sin \theta = s_2 k \rightarrow (1.69)$$

∴ The path difference $s_2 k = (e+d) \sin \theta$ and corresponding phase difference

$$\delta = \frac{2\pi}{\lambda} (e+d) \sin \theta \rightarrow (1.70)$$

To find out resultant amplitude at P_1 we use vector addition method (as shown in figure) in which two sides of a triangle are represented by the amplitudes of s_1 and s_2 and third side gives the resultant amplitude.

According to vector addition method,

From the figure, $(OH)^2 = (OG)^2 + (GH)^2 + 2(OG)(GH) \cos \delta$

$$\begin{aligned} R^2 &= A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 + A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 + 2A \left(\frac{\sin \alpha}{\alpha} \right) A \left(\frac{\sin \alpha}{\alpha} \right) \cos \delta \\ &= 2A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 + 2A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos \delta \\ &= 2A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 [1 + \cos \delta] \\ &= 2A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 2 \cos^2 \frac{\delta}{2} \\ &= 4A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \frac{\delta}{2} \\ &= 4A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \left(\frac{2\pi(e+d) \sin \theta}{\lambda} \right) = 4A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \frac{\pi(e+d) \sin \theta}{\lambda} \end{aligned}$$



$$\rightarrow (1.71)$$

$$\text{Where } \beta = \frac{\pi}{\lambda} (e+d) \sin \theta$$

The above equation (1.71) represents the resultant amplitude of the wavefronts S_1 and S_2 at P_1 .

And intensity $I = R^2$



$$\rightarrow (1.72)$$

From the above equation (1.72), we can say that the resultant intensity (I) at P_1 is the product of two factors

(i) $A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$ which represents the diffraction pattern due to a single slit.

(ii) $\cos^2 \beta$ which represents the interference pattern due to the superposition of secondary diffracted wavelets from two slits s_1 and s_2 .

Therefore, the resultant intensity is the combination of both diffraction and interference effects.

Diffraction effect

In the above equation (1.72), $A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$ gives the principle maximum at the centre of

diffraction pattern with alternative minima and secondary maxima positions on the either side of it (Fig. 1.18).

Case (i) The principle maximum occurs at $\alpha = 0$

$$\text{i.e., } \frac{\sin \alpha}{\alpha} = 1 \quad \rightarrow (1.73)$$

Case (ii) The minimum intensity position occurs at

$$\alpha = \pm m\pi, \text{ where } m = 1, 2, 3, 4, \dots$$

$$\frac{\pi \sin \theta}{\lambda} = \pm m\pi$$

$$\sin \theta = \pm \frac{m\lambda}{a} \quad \rightarrow (1.74)$$

Case (iii) These secondary maxima positions occur at

$$\alpha = \pm (2m+1) \frac{\pi}{2}, \text{ where } m = 1, 2, 3, 4, \dots$$

$$\frac{\pi \sin \theta}{\lambda} = \pm (2m+1) \frac{\pi}{2}$$

$$\sin \theta = \pm (2m+1) \frac{\lambda}{2a} \quad \rightarrow (1.75)$$

Interference effect

In the above equation (1.72), $\cos^2 \beta$ gives the alternative maximum and minimum intensity positions with the equal magnitude of intensity. Case

(i) The Minimum intensity positions will occur at

$$\cos^2 \beta = 0 \text{ When } \beta = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots$$

$$\therefore \beta = \pm (2m+1) \frac{\pi}{2} \rightarrow (1.76)$$

$$\text{where } m = 0, 1, 2, 3, 4, \dots$$

$$\text{But } \beta = \frac{\pi(e+d)\sin\theta}{\lambda} \rightarrow (1.77)$$

From the equations (1.76) & (1.77)

$$\frac{\pi(e+d)\sin\theta}{\lambda} = \pm (2m+1) \frac{\pi}{2}$$

$$\left(\frac{(2m+1)\lambda}{2(e+d)\sin\theta} \right) = \pm \frac{1}{2}$$

→ (1.78)

Case (ii) The maximum intensity positions will occur at

$$\cos^2\beta = 1 \text{ When } \beta = 0, \pm\pi, \pm2\pi, \pm3\pi, \pm4\pi, \dots, \pm m\pi$$

$$\therefore \beta = \pm m\pi \rightarrow (1.79)$$

$$\text{But } \beta = \frac{\pi(e+d)\sin\theta}{\lambda} \rightarrow (1.80)$$

From the equations (1.79) & (1.80)

$$\frac{\pi(e+d)\sin\theta}{\lambda} = \pm m\pi$$

$$\left(\frac{\pi(e+d)\sin\theta}{\lambda} \right) = \pm m\pi$$

→ (1.81)

Intensity distribution graph

Fig 1.18 shows the resultant intensity distribution due to the combination of both diffraction and interference effects. From the graph, it is observed that the resultant minima are not equal to zero; still they have some minimum intensity due to interference effect.

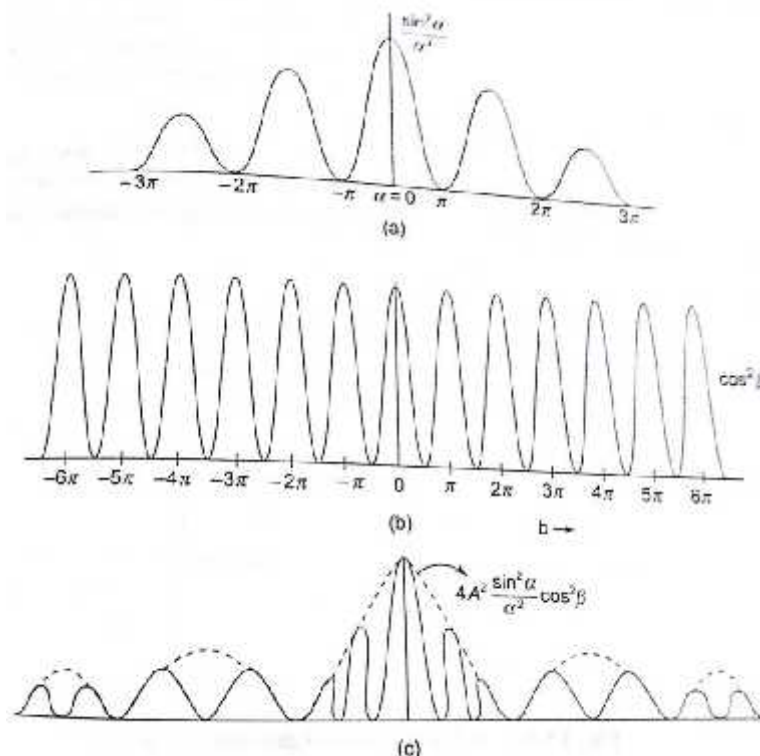


Fig.1.18.Intensity distribution due to diffraction and double slit

2.7. Diffraction Grating

A diffraction grating is an optical device consisting of a large number of equidistant narrow parallel, rectangular slits of equal width and separated by opaque portions. This was first constructed by the German physicist Joseph von Fraunhofer in 1821.

Construction

- A diffraction grating is prepared by ruling a large number of equidistant narrow, parallel lines on an optically plane glass plate with a fine diamond point.
- The ruled lines are opaque to light called opaque portions while the space between any two successive ruled lines is transparent to light called transparent portions and act as slits.
- The distance between any two successive slits (or) ruled lines is called grating element as shown in Fig. 1.19.

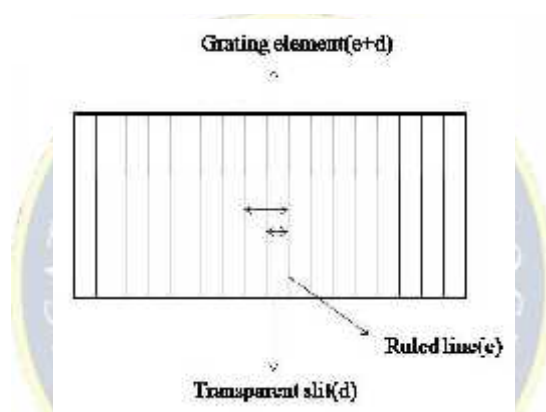


Fig. 1.19. Construction of diffraction grating

- The Commercial gratings are made up a thin layer of colloidal solution (ex: Solution of cellulose Acetate) is poured on the ruled surface and allowed to dry to form a thin film.
- The colloidal thin film is peeled carefully from the ruled surface. The film retains an impression of the ruling of ruled surface.
- The ruled lines act as opaque portions whereas the space between them act as transparent portions which transmit incident light. The film is mounted between two glass plates called grating.
- When the light is incident on the grating surface then light is transmitted through transmission portions or slits and obstructed by the opaque portions or ruled lines. Such a grating is known as a transmission grating.

Theory

Let 'e' be the width of each ruled line (or) opaque portion and 'd' be the width of transparent portion (or) slit then (e + d) represents the grating element. The relation between grating element and the angles of diffracted beams of light is known as grating equation. It is given by

$$(e+d)\sin\theta=n\lambda, \text{ Where } n=0,1,2,3,\dots$$

→(1.82)

This expression is known as grating equation.

$$\sin\theta=\frac{n\lambda}{e+d}$$

$$\sin\theta=Nn\lambda$$

Where $\frac{1}{e+d} = N$ is the number of grating elements or lines per unit width of the grating.

$$N(e+d)=1 \text{ inch} = 2.54 \text{ cm}$$

$$e+d = \frac{2.54}{N} \text{ cm}$$

→(1.83)

2.8. Grating Spectrum

The diffraction pattern formed with a grating is known as grating spectrum.

The position of principal maxima in grating are given by

$$(e+d)\sin\theta=n\lambda, \text{ Where } n=0, 1, 2, 3,\dots$$

This expression is known as grating equation.

For a particular wavelength, the diffraction angle is different for different principal maxima of different orders.

The first order maximum is obtained for $n=1$, then $(e+d)\sin\theta_1=\lambda$

The second order maximum is obtained for $n=2$, then $(e+d)\sin\theta_2=2\lambda$

The different orders of principal maxima are obtained on both sides of zero order maxima as shown in Fig. 1.20.

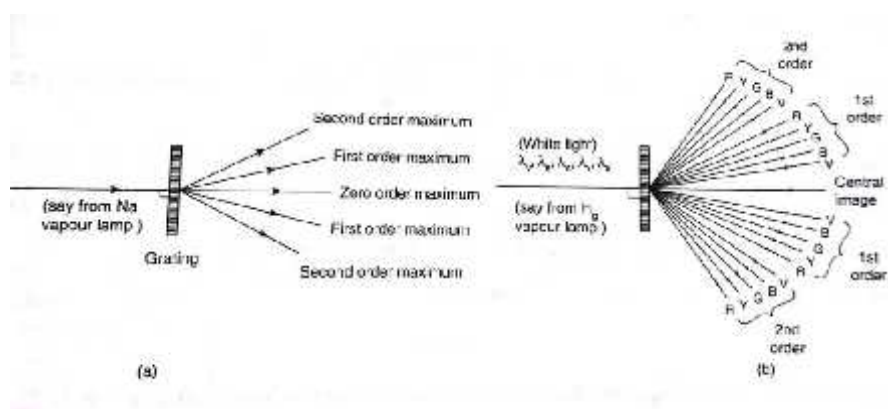


Fig.1.20. Grating spectrum (a) for monochromatic source (b) for white light source

2.9. Determination of Wavelength (λ) of given source of light

The positions of principle maxima in grating are given by

$$(e+d)\sin\theta = n\lambda, \text{ Where } n = 0, 1, 2, 3, \dots$$

$$\sin\theta = \left(\frac{1}{e+d}\right)n\lambda$$

$$\sin\theta = N\lambda$$

$$n\lambda = \frac{\sin\theta}{N} \quad \rightarrow (1.84)$$

Where $\frac{1}{e+d} = N$ is the number of grating elements or lines per unit width of the grating.

By knowing the values of angle of diffraction, order of the principal maxima and grating element, the wavelength can be determined.

2.10. Applications of diffraction

The phenomenon of diffraction can be used in various engineering applications; some of the applications are given below.

1. It is used in x-ray diffraction studies of crystals.
2. It is also used in holography for reconstructing 3D images of objects using laser light.
3. Diffraction gratings are used in spectroscopes to separate a light source into its component wavelengths.
4. Diffraction grating can be chosen to specifically analyze a wavelength of light emitted by molecules in diseased cells in a biopsy sample.
5. Diffraction gratings are used in optical fiber technologies where fibers are designed to provide optimum performance at specific wavelengths.

2.11. POLARIZATION

2.11.1 Introduction

The phenomena of interference and diffraction tells us that light is a form of wave. But they do not tell us whether light is transverse or longitudinal wave in nature. The phenomenon of polarization confirms that light waves are transverse in nature.

In general, light waves are electromagnetic waves and contain vibrations of electric and magnetic field vectors (E & B) which are perpendicular to each other and also perpendicular to the direction of propagation of light waves .i.e., transverse waves. Therefore, light is a transverse wave motion. It has been found experimentally that the electric field vector (E) component of light is mainly responsible for polarization and other optical effects but not due to the magnetic field vector.

Therefore, the electric field vector E is also called the light vector and vibration of light means vibration of electric field vector of light.

The word polarization comes from the Greek word “polos” which means orientation. Therefore the phenomenon of restricting or orienting the vibration of the electric field vector in a particular direction is called polarization of light.

2.11.2. Unpolarized light

If the light vector (electric field vector E) vibrates in all possible directions which are perpendicular to the direction of propagation then the light is said to be unpolarized light.

Let us consider an ordinary light ray passing perpendicular to the plane of the paper and into the paper. The electric field vectors are perpendicular to the ray propagating with equal amplitude in all possible directions as shown in Fig. 1.21.

This is the nature of unpolarized light. A double-headed arrow shows the to and fro vibration of the electric vector.

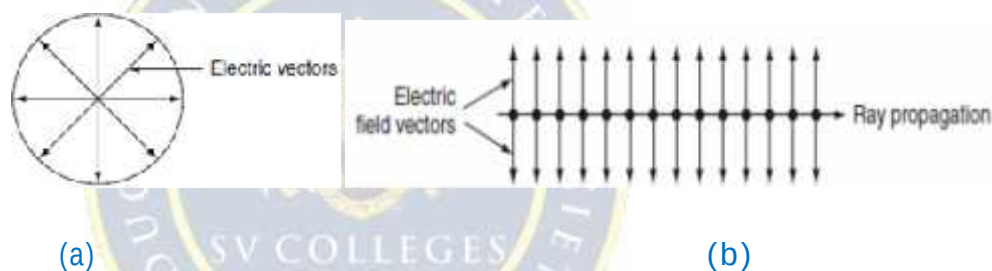


Fig 1.21. Representation of unpolarized light

2.12. Polarized light

The phenomenon of restricting the vibration of the light vector or electric field vector (E) in a particular direction is called polarization of light. The polarized light can be classified into three types. They are

- i. Linearly (or) Plane polarized light
- ii. Circularly polarized light and
- iii. Elliptically polarized light

Linearly (or) Plane polarized light

If the vibration of light vector (electric field vector E) is restricted along a straight line only in a plane perpendicular to the direction of propagation of light, it is said to be plane polarized light.

The linearly polarized light is shown in 1.22. The arrowed lines represent in the plane of paper and the dots, the vibrations at right angles (perpendicular) to the paper. In Fig.1.22(a), the vibrations of electric field vectors lie in the plane of the paper and in Fig.1.22(b) the vibrations of electric field vectors are perpendicular to the plane of the paper.

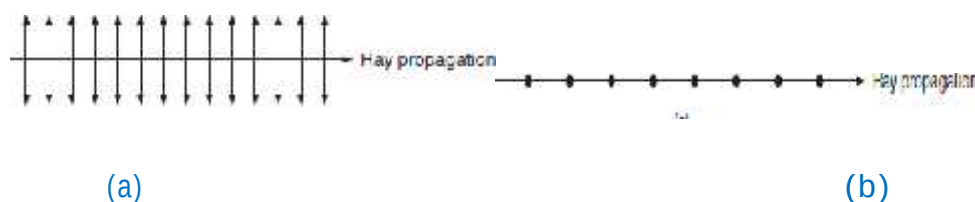


Fig1.22.Representation of plane polarized light

Circularly polarized light and elliptically polarized light

When two plane polarized light waves are superimposed on each other then under certain conditions the resultant light vector may rotate with a constant magnitude in a plane perpendicular to the direction of propagation of light.

If the magnitude of the resultant light vector remains constant while its orientation varies regularly, the tip of the vector traces a circle. Thus the light is said to be circularly polarized. If however, both magnitude and orientation of light vector vary, the tip of the vector traces an ellipse. Thus the light is said to be elliptically polarized.

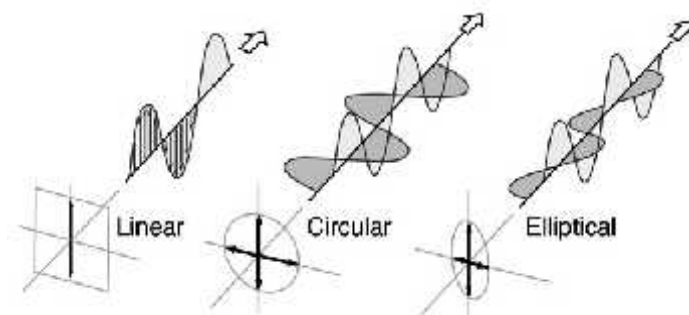


Fig1.23.Representation of plane, circular and elliptically polarized light

2.13. Methods of producing polarized light

Polarized light can be produced by any one of the following.

- Polarization by reflection
- Polarization by refraction
- Polarization by double refraction
- Polarization by scattering
- Polarization by selective absorption

In the present section we only discuss only polarization by double refraction.

2.14. Polarization by double refraction

When a light ray is passed through a glass plate, we get only one refracted ray. But when a light ray is passed through a certain crystal like calcite or quartz, we get two refracted rays, one refracted light ray which obeys the laws of refraction is called ordinary or o-ray. The other ray does not obey the laws of refraction and is extraordinary ray or e-ray. This phenomenon is called double refraction or birefringence. This was first discovered by Erasmus Bartholinus in 1669.

Explanation

When a ink dot is marked on a white paper and is seen through a calcite crystal, then two images (dots) are observed. If the crystal is rotated slowly with incident ray as vertical axis, then it is observed that one image remains fixed and the other image rotates with the rotation of the calcite crystal. The fixed image is called ordinary image and its refracted ray is called o-ray which has vibrations perpendicular to plane of the white paper. The other image is called extraordinary image and its refracted ray is called e-ray which has vibrations in the plane of the paper as shown in Fig. 1.24.

Inside the crystal the o-ray travels with the same velocity in all direction.

But the e-ray has different velocities in different directions. If the incident ray strikes the crystal along a certain direction called the optical axis of the crystal, there will be no double refraction. Hence optical axis of the crystal is a direction along which both o-ray and e-ray travel in the same direction with the same velocity.

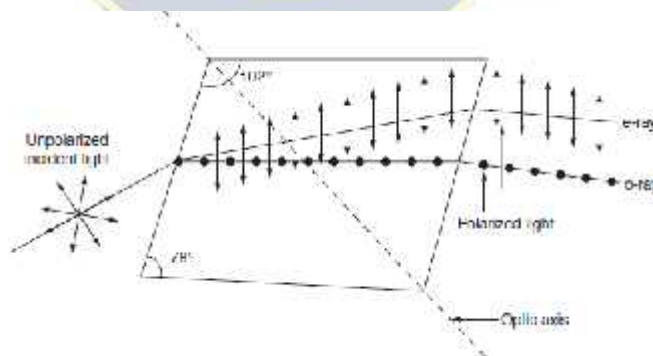


Fig.1.24. Double refraction in calcite crystal

2.15. Double refracting crystals

The crystals which exhibit double refraction are called double refracting crystals.

There are two types of double refracting crystals

- i. uni-axial crystals
- ii. biaxial crystals

Uni-axial crystals

If only one optic axis is present in the crystals, then they are called uni-axial crystals. Examples: Calcite, quartz and tourmaline etc.

These crystals further divide into two crystals

- i. Uni-axial negative crystals
- ii. Uni-axial positive crystals

Uni-axial negative crystals

If the velocity of e-ray is more than that of o-ray in the crystals, then they are called negative crystals. i.e., $v_e > v_o$ and $\mu_e < \mu_o$

Examples: Calcite, tourmaline, ruby and emerald etc

Uni-axial positive crystals

If the velocity of e-ray is less than that of o-ray in the crystals, then they are called positive crystals. i.e., $v_e < v_o$ and

$\mu_e > \mu_o$ Examples: Quartz and iron oxide. Biaxial crystals

If two optic axes are present in the crystals, then they are called biaxial crystals. Examples: Topaz and aragonite.

Note:

Optical axis

The direction in which the ray of transmitted light does not suffer double refraction inside the crystal is known as the optic axis. Along this axis both the velocities of o-ray and e-ray are the same and also refractive indices are same. i.e., $v_e = v_o$ and $\mu_e = \mu_o$. So, there is no double refraction. Optical axis is a direction but not a line.

2.16. Wave Plates or Retardation Plates

Wave plates are the doubly refracting uniaxial (calcite) crystals whose refracting faces are cut parallel to the direction of the optic axis and are used to produce a phase difference between ordinary and extraordinary rays when they emerge from the doubly refracting crystals.

These are of mainly two types

- i. Quarter-wave plate
- ii. Half-wave plate

2.16.1. Quarter-waveplate

A quarter-

waveplate is a thin double refracting crystal (calcite) having a thickness 't', cut and polished with its refracting faces parallel to the direction of optic axis such that

it produces a path difference of $\lambda/4$ or phase difference of $\pi/2$ between the o-ray and e-

ray when plane polarized light is incident normally on the surface and passes through the plate.

Consider a calcite crystal of thickness of 't'. When a plane polarized light is incident normally on the surface of calcite crystal, then the light will split up into o-ray and e-ray. These rays travel with different velocities in the crystal. As a result, when o-ray and e-

ray are merging from the crystal, they have a phase or path difference between them due to variation in their velocities.

Let μ_o and μ_e be the refractive indices of o-ray and e-ray respectively. Let 't' be the thickness of the crystal. Hence the path difference between the two rays is

$\Delta = \text{Optical path for o-ray} - \text{Optical path for e-ray}$

$$= \mu_o t - \mu_e t = (\mu_o - \mu_e)t \quad \rightarrow (1.85)$$

As the crystal is a quarter-waveplate, it introduces a path difference of $\lambda/4$ between o-ray and e-ray.

$$\Delta = \lambda/4 \quad \rightarrow (1.86)$$

From Eqs. (1.57) and (1.58), we get

$$\begin{aligned} (\mu_o - \mu_e)t &= \frac{\lambda}{4} \\ t &= \frac{\lambda}{4(\mu_o - \mu_e)} \\ t &= \frac{\lambda}{4(\mu_e - \mu_o)} \\ t &= \frac{\lambda}{4(\mu_o - \mu_e)} \end{aligned}$$

Therefore

For positive crystal $\mu_e > \mu_o$, then

For negative crystal $\mu_o > \mu_e$, then

Applications

- A quarter waveplate is used to produce circularly and elliptically polarized light.
- Quarter wave plate converts plane-polarized light into elliptically or circularly polarized light depending upon the angle that the incident light vector makes with the optic axis of the quarter waveplate.

2.16.2. Half-waveplate

A half-wave plate is a thin double refracting crystal (calcite) having a thickness 't', cut and polished with its refracting faces parallel to the direction of optic axis such that it produces a path difference of $\frac{\lambda}{2}$ or phase difference of π between the o-ray and e-ray when plane polarized light is incident normally on the surface and passes through the plate.

As the crystal is a half-wave plate, it introduces a path difference between o-ray of $\frac{\lambda}{2}$ and e-ray.

$$(\mu_e - \mu_o)t = \frac{\lambda}{2}$$

$$t = \frac{\lambda}{2(\mu_e - \mu_o)}$$

$$t = \frac{\lambda}{2(\mu_o - \mu_e)}$$

Therefore

For positive crystal $\mu_e > \mu_o$, then

For negative crystal $\mu_o > \mu_e$, then

Applications

1. A half-wave plate is used to produce plane polarized light.
2. It produces a phase difference of π between the ordinary and extraordinary ray.

2.17. Nicol's Prism

Nicol prism is an optical device used to produce and analyze plane polarized light. This was invented by William Nicol in the year 1828 and is known as Nicol prism. It is made from a double refracting calcite crystal.

Principle

It is based on the phenomenon of double refraction. When a light ray is passed through a calcite crystal, it splits up into two refracted rays such as ordinary (O-ray) and extraordinary ray (E-ray). Nicol prism transmits the extraordinary rays and eliminates ordinary rays with the help of the phenomenon of total internal reflection.

Construction

- It is constructed from the calcite crystal ABCD having length three times of its width.
- The end faces AB and CD are cut down such that the angles of principal section are 68° and 112° instead of 71° and 109° .
- The crystal is then cut diagonally into two parts. The surfaces of these parts are ground to make optically flat and then these are polished.

- These polished surfaces are connecting together with special cement known as Canada balsam which is a transparent liquid material.
- Canada balsam is optically (refractive index) more dense than e-ray and less dense for o-ray ($\mu_o = 1.6584$, $\mu_{ca} = 1.55$ and $\mu_e = 1.4864$).

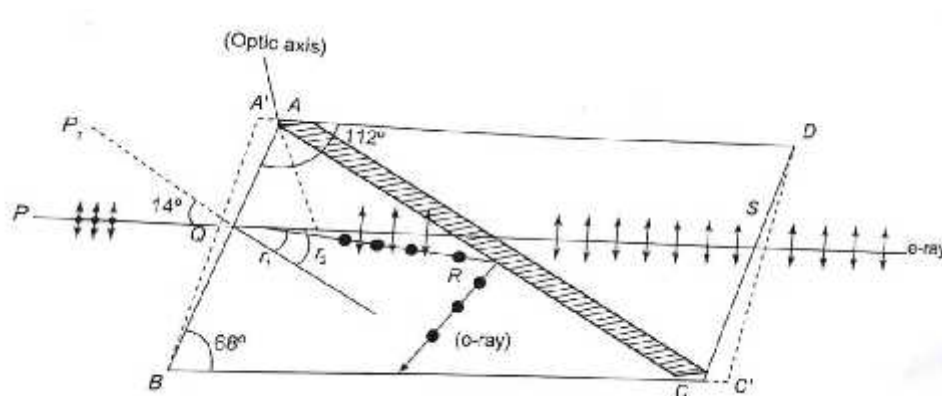


Fig.1.25. Nicol prism

Action or Working

- When a beam of unpolarized light PQ is incident on the face of AB, it splits into two refracted rays: o-ray (QR) and e-ray (QS) due to double refraction of calcite crystal.
- These two rays are plane polarized.
- From the refractive index values, we know that the Canada balsam acts as a rarer medium for the ordinary ray and acts as a denser medium for the extraordinary ray.
- When the o-ray of light travels in the calcite crystal and enters the Canada balsam cement, it passes from denser to rarer medium. When the angle of incidence for the ordinary ray on the Canada balsam is greater than the critical angle, then the incident ordinary ray is totally internally reflected from the crystal and only the e-ray is transmitted through the prism and emerges out of the Nicol prism.
- In this way, plane polarized light is produced.

2.18. Nicol prism as polarizer and analyzer

- Nicol prism can be used as a polarizer and analyzer.
- In order to produce and analyze plane polarized light, two Nicol prisms are arranged adjacently as shown in Fig.1.26.
- The first Nicol prism is used to produce plane polarized light and is called a polarizer. The second Nicol prism is used to test the emerging light and is called an analyzer.

- In the parallel positions, the extraordinary ray passes through both the prisms as shown in Fig.1.29 (a). In this case, the intensity of emergent extraordinary light is maximum.
- If the second prism is slowly rotated, then the intensity of the extraordinary ray decreases. When they are perpendicular to each other, no light comes out of the second prism because the e-ray that comes out from first prism will enter into the second prism and act as an ordinary ray. So, this light is reflected in the second prism as shown in Fig.1.29(b).

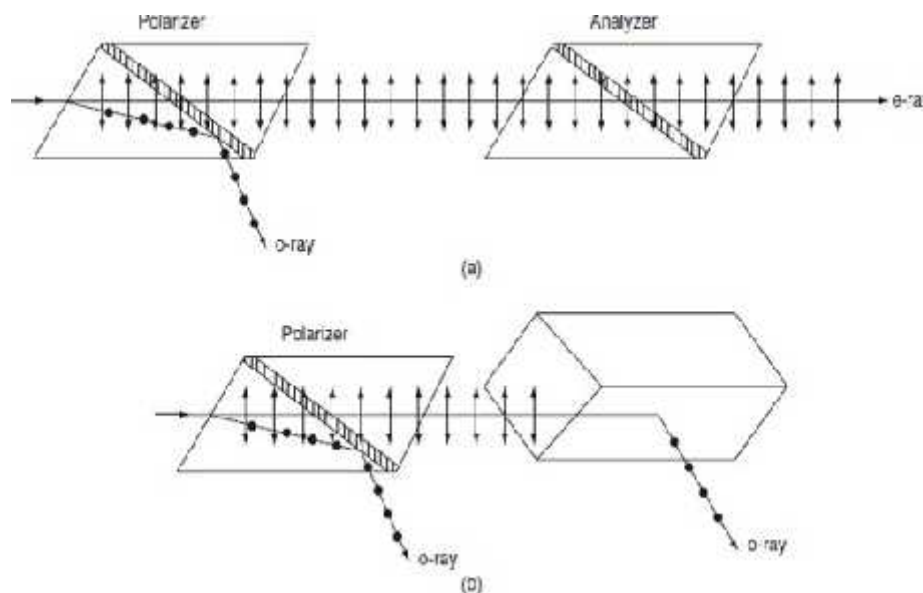


Fig.1.26.Nicol prism as polarizer and analyzer

2.19. Applications of Polarization

The phenomenon of Polarization can be used in various engineering applications; some of the applications are given below.

1. Polaroid glasses are used to reduce the amount of light that is approachable to eye.
2. Polarization is useful in receiving and transmitting wave signals.
3. Laser is an outcome of polarization of waves.
4. 3D movies (or stereoscopic movies) are possible because of polarization of light.
5. Photographic filters.
6. Photoelasticity – To study the objects with irregular boundaries and stress analysis.
7. LCD's.

9. Practice Quiz

- When the light wave is reflected at the air-glass interface, the change of phase of the reflected wave is equal to [A] (a) 0
(b) $\pi/2$ (c) $\pi/4$ (d) π
- Two beams interfere having their amplitudes ratio 2:1. Then the intensity ratio of bright and dark fringes is [D]
(a) 2:1 (b) 1:2 (c) 9:1 (d) 4:1
- In Newton's rings interference is due to light rays reflected from [B]
(a) Lower surface of lens and upper surface of glass plate
(b) Lower surface of glass plate and upper surface of lens
(c) Lower surface of lens and lower surface of glass plate
(d) Upper surface of lens and upper surface of glass plate
- The penetration of waves into the region of the geometrical shadow is [B]
(a) interference (b) diffraction (c) polarization (d) dispersion
- In single slit diffraction, the first diffraction minima is observed at an angle of 30° , when the light of wavelength 500 nm is used. The width of the slit is [C] (a) 5×10^{-5} cm (b) 2.5×10^{-5} cm (c) 10×10^{-5} cm (d) 1.25×10^{-5} cm
- In a single slit experiment if the slit width is reduced [A]
(a) The fringes become brighter
(b) The fringes become narrower
(c) The fringes become wider
(d) The colour of the fringes change
- Instead of red colour source, if blue colour source is used in single slit experiment [C]
(a) The diffraction pattern does not change
(b) The diffraction bands become wider
(c) The diffraction pattern becomes narrower and crowded together
(d) The diffraction pattern disappears

FILL IN THE BLANKS

- Due to interference in thin films, if the film thickness is extremely small when compared to λ , the film will appear DARK
- In grating, the combined width of a ruling and a slit is called GRATING ELEMENT
- The phenomenon which confirms the transverse nature of light is POLARIZATION

10. Assignment Questions:

S.No	Question	BL	CO
1	Define interference? Explain interference in thin films by reflection of light.	1	1
2	What are Newton rings? Explain theory of Newton rings.	1	1
3	Mention the expression for diameter of bright and dark rings in Newton's rings by the reflected light.	3	1
4	Explain Fraunhofer diffraction due to double slit.	2	1
5	Define Plane polarization. Describe the construction and working of Nicol prism.	1	1
6	Distinguish between interference and diffraction.	2	1

11. Part A-Question & Answers

S.No	Question & Answers	BL	CO
1	What is meant by optics? Answer: Optics is the branch of physics, which deals with nature, properties, generation, and propagation of light and its interaction with matter.	1	1
2	What is interference? Answer: The formation of bright and dark fringes on superimposing two coherent light waves is called interference.	1	1
3	What are Newton rings? Answer: When a Plano-convex lens with its convex surface is placed on a plain glass plate, then an air film is formed between the lower surface of the Plano-convex lens and the upper surface of the glass plate. If a monochromatic light is allowed to fall normally on this air film, a system of alternate bright and dark circular rings is formed in the air film. These rings were discovered by Newton and are called Newton's rings.	1	1
4	What is the principle of Newton rings? Answer: Newton's rings are formed due to interference between the light rays reflected from the top and bottom surfaces of an air film between the glass plate and the Plano-convex lens.	1	1

5	When sunlight falls on the oil layers and soap bubbles, different colors are produced, why? Answer: When a sun light falls on the soap bubbles, a film of oil floating over a water, butterfly wings and peacock feathers different colors are seen, this is due to the interference (resulting from the light between the two surfaces of the thin films) in thin films by reflection of light. This was first explained by Thomas Young on the basis of interference.	1	1
6	What is diffraction of light? Write types of diffraction? Answer: The bending of light waves at the corners or edges of obstacles or slits is called diffraction. There are two types of diffractions a) Fraunhofer diffraction b) Fresnel diffraction.	1	1

12. Part B-Questions

S.No	Question	BL	CO
1	What is interference? Explain interference in thin films by reflection of light.	1	1
2	What are Newton rings? Explain theory of Newton rings.	1	1
3	Newton rings are observed in the reflected light of wavelength $5900 \text{ \AA}$. The diameter of 10th dark ring is 0.5 cm . Find the radius of curvature of lens used?	1	1
4	Explain Fraunhofer diffraction due to double slit.	2	1
5	Explain with necessary theory how wavelength of spectral line is determined using plane diffraction grating.	2	1

13. Supportive Online Certification Courses

1. Engineering Physics I (Experimental) by Prof. Alike Khare Prof. Pratima Agarwal Prof. S. Raviconducted by IIT Guwahati-12 weeks
2. Engineering Physics I (Theory) by Prof. Alike Khare Prof. Pratima Agarwal Prof. S. Raviconducted by IIT Guwahati-12 weeks

14. Prescribed Text Books & Reference Books Text Book

1. A text book of Engineering Physics –
Dr. M. N. Avadhanulu & Dr. P. G. Kshirsagar,
S. Chand and Company, 11th Edition, 2019.
2. Engineering Physics – B. K. Pandey and S. Chaturvedi, Cengage Learning, 2013.

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